

EXPERIMENT 10 : Transformation-Based Tagging

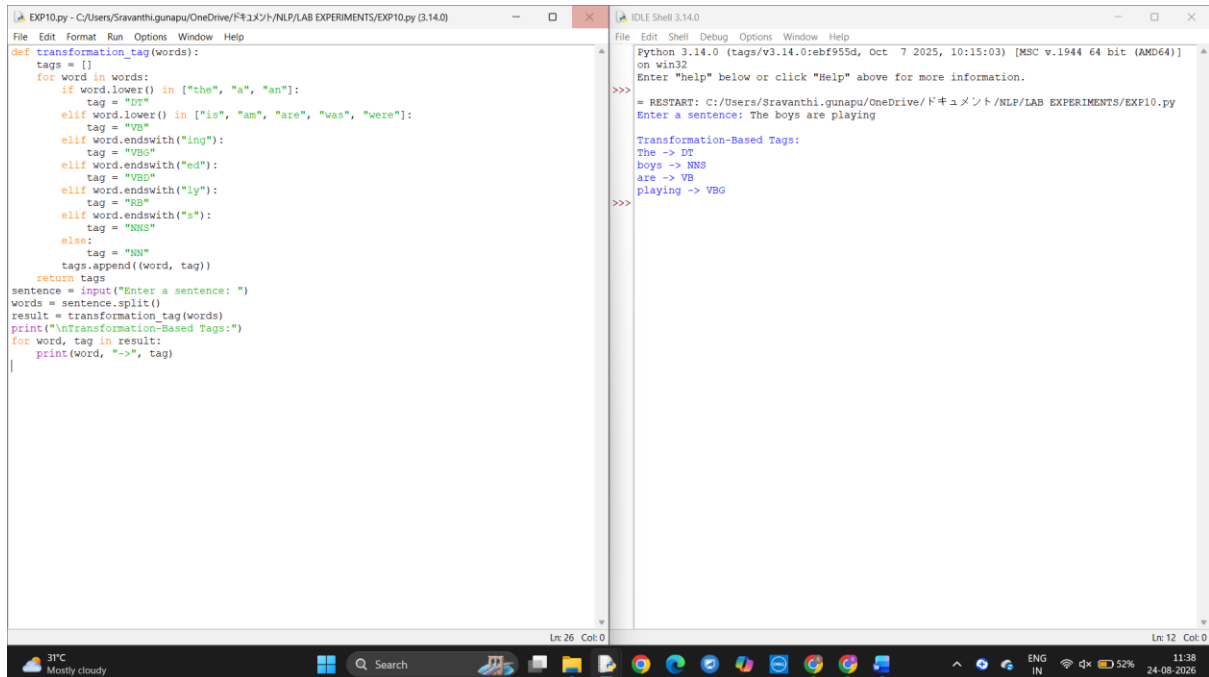
Aim: To implement transformation-based tagging using a set of transformation rules in Python.

Program:

```
def transformation_tag(words):
    tags = []
    for word in words:
        if word.lower() in ["the", "a", "an"]:
            tag = "DT"
        elif word.lower() in ["is", "am", "are", "was", "were"]:
            tag = "VB"
        elif word.endswith("ing"):
            tag = "VBG"
        elif word.endswith("ed"):
            tag = "VBD"
        elif word.endswith("ly"):
            tag = "RB"
        elif word.endswith("s"):
            tag = "NNS"
        else:
            tag = "NN"
        tags.append((word, tag))
    return tags

sentence = input("Enter a sentence: ")
words = sentence.split()
result = transformation_tag(words)
print("\nTransformation-Based Tags:")
for word, tag in result:
    print(word, "->", tag)
```

Output:



The image shows a screenshot of a Python IDE with two windows. The left window, titled 'EXP10.py', contains a Python script for word tagging. The script defines a function 'transformation_tag' that takes a list of words and returns a list of tags. It uses a series of 'if' and 'elif' statements to assign tags based on word endings. For example, words ending in 'the', 'a', or 'an' are tagged 'DT'; words ending in 'is', 'am', 'are', 'was', or 'were' are tagged 'VB'; words ending in 'ing' are tagged 'VBG'; words ending in 'ed' are tagged 'VBD'; words ending in 'ly' are tagged 'RB'; words ending in 's' are tagged 'NNS'; and all other words are tagged 'NN'. The script then prompts the user to enter a sentence, splits it into words, and prints the transformation-based tags for each word.

```
def transformation_tag(words):
    tags = []
    for word in words:
        if word.lower() in ["the", "a", "an"]:
            tag = "DT"
        elif word.lower() in ["is", "am", "are", "was", "were"]:
            tag = "VB"
        elif word.endswith("ing"):
            tag = "VBG"
        elif word.endswith("ed"):
            tag = "VBD"
        elif word.endswith("ly"):
            tag = "RB"
        elif word.endswith("s"):
            tag = "NNS"
        else:
            tag = "NN"
        tags.append((word, tag))
    return tags
sentence = input("Enter a sentence: ")
words = sentence.split()
result = transformation_tag(words)
print("\nTransformation-Based Tags:")
for word, tag in result:
    print(word, "->", tag)
```

The right window, titled 'IDLE Shell 3.14.0', shows the output of the script. It displays the Python version (3.14.0), the file path, and the prompt 'Enter "help" below or click "Help" above for more information.' Below this, it shows the restart command and the input sentence 'The boys are playing'. The output then lists the transformation-based tags for each word: 'The -> DT', 'boys -> NNS', 'are -> VB', and 'Playing -> VBG'.

```
Python 3.14.0 (tags/v3.14.0:ebf955d, Oct 7 2025, 10:15:03) [MSC v.1944 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.

>>>
= RESTART: C:/Users/Sravanthi.gunapu/OneDrive/ドキュメント/NLP/LAB EXPERIMENTS/EXP10.py
Enter a sentence: The boys are playing

Transformation-Based Tags:
The -> DT
boys -> NNS
are -> VB
Playing -> VBG

>>>
```

The Windows taskbar at the bottom shows the system clock as 11:38 on 24-08-2026, with a battery level of 52% and network status.